



Department of Information Technology
Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT

Course Code & Title: CCS354 & Network Security

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / IP Security , Internet Key Exchange
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** One Minute Paper
- **Justification:**

IP Security (IPsec) is a suite of protocols ensuring secure, private IP communications by authenticating and encrypting packets. Internet Key Exchange (IKE) automates this by negotiating security associations (SAs) and generating keys via Diffie-Hellman, enabling secure VPN connections and managing key exchanges for IPsec across untrusted networks. The one-minute paper is a rapid, formative assessment tool where students spend one minute at the end of class which provides immediate feedback on comprehension, encouragement of active reflection, improved retention.

- **Time Allotted for the Activity:** 01 Minute
- **Details of the Implementation:**
 - At the end of the lecture, students were given two topics IP Security and Internet Key Exchange (IKE).
 - Students were asked to write about the given topics for a minute. Few students chosen the topic IP Security and few of them chosen Internet Key Exchange.
 - Fig 1a shows the One Minute paper sample of K.Gayathri , where she written about IP
 - Fig 1b shows the One Minute paper sample of P.Saranya , where she has written about Internet Key Exchange.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	1	2	1	1	2	3	3	1	3	2	1	3

(1 – Low

2 – Moderate

3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	1	2	1	1	2	3	3	1	3	2	1	3
Justification for Correlation	Recognizing effective network security measures requires a deep understanding of network engineering principles to identify critical vulnerabilities and potential attack points	Effective network security measure identification involves analyzing system requirements and potential threats to pinpoint areas susceptible to security vulnerabilities	Identifying effective network security measures involves designing security solutions that target critical vulnerabilities and potential attack vectors, contributing to a robust and secure network environment	Identifying effective network security measures requires the ability to investigate complex network behaviors and dependencies to uncover potential security vulnerabilities	Leveraging modern security tools and techniques aids in identifying effective network security measures by enhancing coverage and efficiency in detecting vulnerabilities and threats.	Effective network security measure identification involves collaboration within teams and individual contributions to brainstorm, prioritize, and refine security strategies to uncover critical vulnerabilities	Communicating the importance of identifying effective network security measures to stakeholders and team members ensures alignment and support for security initiatives	Identifying effective network security measures involves considering resource allocation, timelines, and project constraints to prioritize security activities effectively	Recognizing the importance of identifying effective network security measures requires a commitment to continuous learning to stay updated with evolving threats and technologies	Applying effective network security measures demonstrates proficiency in network engineering, ensuring the identification and resolution of complex security challenges	Identifying effective network security measures helps develop skills in emerging technologies such as mobile application security, ensuring the delivery of reliable and secure IT solutions	Carrying effective network security measures showcases strategies in securing network infrastructures using open-source tools to deliver reliable and secure solutions for business and industry needs.

- Images / Screenshot of the practice:

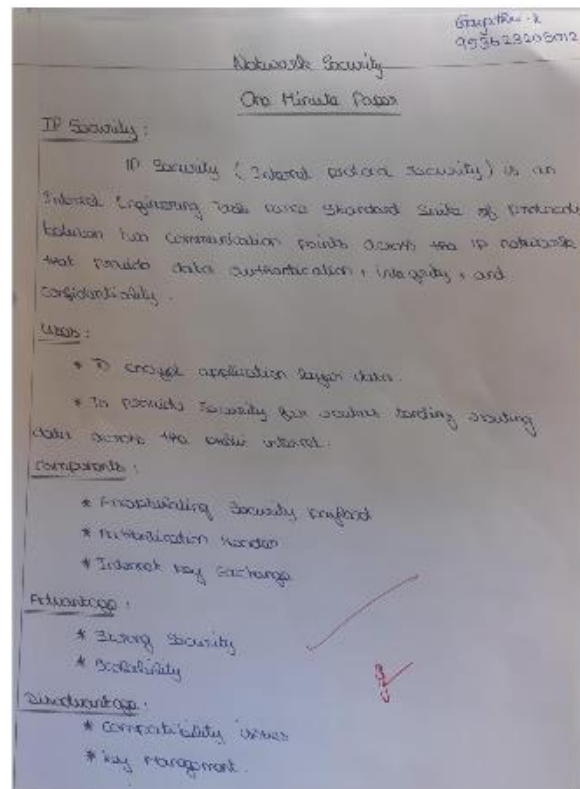


Figure 1. a Sample One Minute Paper by K.Gayathri

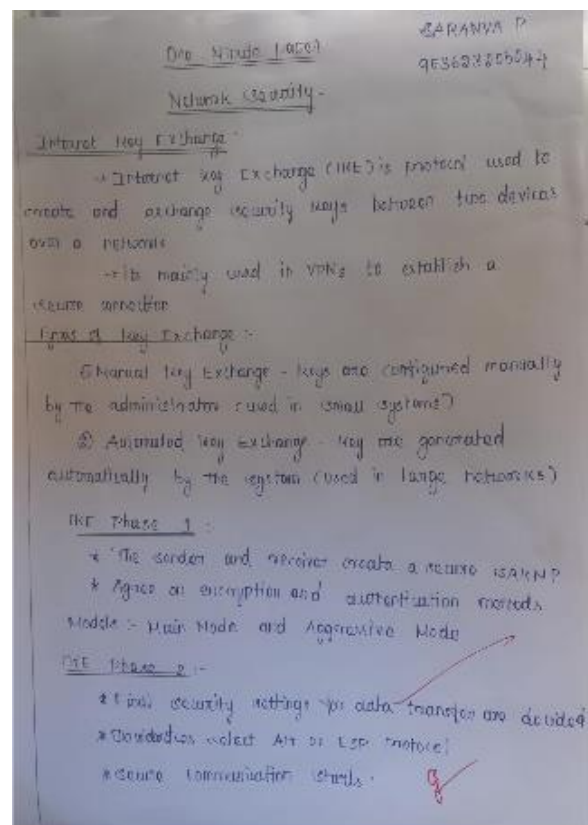


Figure 1. b Sample One Minute Paper by P.Saranya

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students reported that the one-minute paper helped them quickly organize their thoughts at the end of the class.
- Some stakeholders noted that it encouraged quieter students to participate indirectly.

❖ *Benefit of the practice:*

- Students actively reflected on what they learned, improving retention.
- It promotes metacognitive thinking by making students evaluate their own understanding.
- It enhances writing clarity and concise expression skills.
- It supports inclusive learning by giving every student an equal voice.

❖ *Challenges faced in implementation:*

- Reviewing responses regularly can be time-consuming for educators in large classes.
- Students unfamiliar with reflective learning may initially struggle to articulate their thoughts effectively.

❖ *References:*

- <https://teaching.unl.edu/resources/interactive-lecture/minute-paper/>
- <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- <https://www.eur.nl/en/educational-innovation/teacheur/one-minute-paper>
- <https://mtei.engineering.cornell.edu/teaching-resources/active-learning/one-minute-papers/>

Signature of Faculty Member

HOD